BRIEF TEKNIS

DigiVote.id

Platform E-Voting Berbayar · Panduan Pengembangan

Laravel · MySQL · Hostinger Cloud Professional · Xendit QRIS

Versi 1.0 | Untuk Tim Developer Internal | 2025

Daftar Isi

1. Gambaran Umum Produk
2. Arsitektur Sistem
3. Struktur Modul & Fitur
4. Rancangan Database (MySQL)
5. Optimasi Performa — Hostinger Cloud Pro
6. Integrasi Xendit QRIS
7. Struktur Direktori Laravel
8. Prioritas Development & Timeline

1. Gambaran Umum Produk

1.1 Apa itu DigiVote.id?

DigiVote.id adalah platform e-voting berbayar berbasis web yang memungkinkan penyelenggara (organizer) membuat event voting, menambahkan tim peserta, dan membagikan link/QR code kepada pendukung. Setiap vote diproses melalui pembayaran QRIS real-time via Xendit.

1.2 Alur Utama Pengguna

Aktor	Alur Aksi	Endpoint Utama
Organizer / Super Admin	Buat event ⟶ assign admin event → pantau pemasukan	/admin/*
Admin Event	Login dengan ID event → tambah kategori & tim → pantau leaderboard, votes, pemasukan → download QR	/admin-event/*
Voter (Publik)	Buka link/scan QR → pilih tim → input qty → bayar QRIS → vote tercatat	/{slug}?kandidat={no}

1.3 Constraint Hosting (Hostinger Cloud Professional)

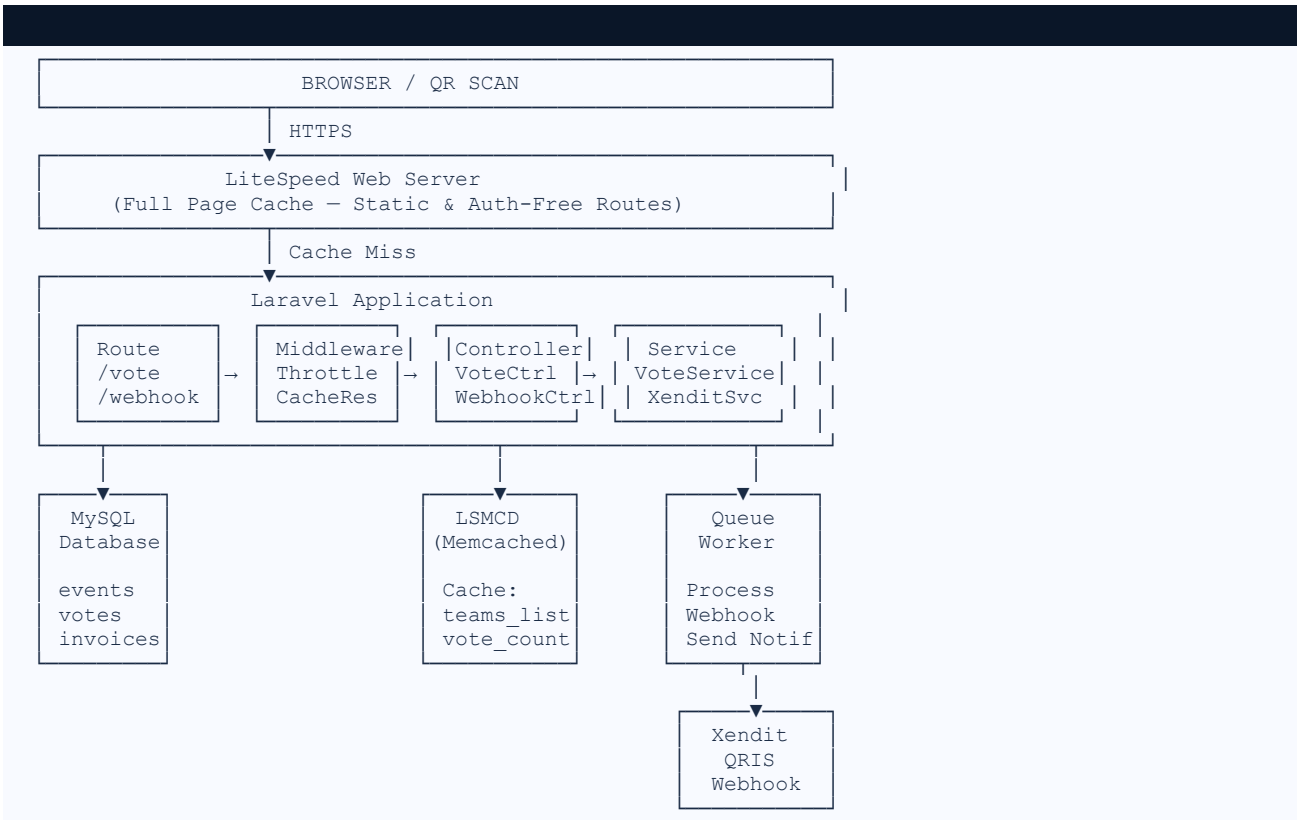
Spesifikasi	Nilai	Implikasi Pengembangan
PHP Worker	200	Max 200 request Laravel bersamaan — wajib gunakan caching agresif
CPU Core	5 Core (shared)	Hindari proses berat di request cycle — gunakan Queue
RAM	6 GB (shared)	Batasi memory per proses, optimalkan query
NVMe Storage	200 GB	Aman untuk media dan storage
Object Cache	LSMCD (Memcached)	Set CACHE_DRIVER & SESSION_DRIVER=memcached
Web Server	LiteSpeed	Full page cache otomatis, tidak perlu config ekstra
Database	MySQL (shared)	Wajib index pada kolom yang sering di-query

KRITIS: 10.000 concurrent user adalah target. Strategi caching agresif WAJIB diimplementasi dari awal, bukan setelah deployment.

2. Arsitektur Sistem

2.1 Arsitektur High-Level

DigiVote.id menggunakan arsitektur monolitik Laravel dengan pemisahan concern yang jelas melalui Service Layer. Tidak digunakan microservice untuk menjaga simplisitas di infrastruktur shared hosting.



2.2 Komponen Utama

Komponen	Teknologi	Fungsi
Web Server	LiteSpeed (Hostinger)	Full page cache otomatis, HTTP/3
Framework	Laravel 12.x	MVC + Service Layer + Queue
Database	MySQL 8.0	Penyimpanan data utama
Object Cache	LSMCD (Memcached)	Cache query, session, rate limiter
Queue Driver	Database / Redis (opsional)	Async job processing
Payment Gateway	Xendit QRIS	Invoice creation, webhook handler
QR Generator	api.qrserver.com atau library PHP	QR code per kandidat
File Storage	Local NVMe	Asset & export file

3. Struktur Modul & Fitur

3.1 Modul Public (Voter — Tanpa Login)

Route	Method	Fungsi	Cache Strategy
/{slug}	GET	Halaman event + daftar tim	Full page cache LiteSpeed
/{slug}?kandidat={no}	GET	Buka langsung form vote kandidat	Full page cache
/vote/{eventSlug}	POST	Submit form vote → buat invoice Xendit	No cache, throttle 60/min/IP
/webhook/xendit	POST	Terima konfirmasi bayar dari Xendit	No cache, queue dispatch
/qr/{eventSlug}/{teamNum}	GET	Redirect ke form vote kandidat	Cache 24 jam

3.2 Modul Admin Event (Login per Event)

Route	Method	Fungsi
/admin-event/login	GET/POST	Login dengan Event ID + password
/admin-event/overview	GET	Dashboard: votes, income, status
/admin-event/leaderboard	GET	Ranking tim real-time
/admin-event/income	GET	Rincian pemasukan per tim
/admin-event/votes	GET	Tabel detail votes
/admin-event/qr-links	GET	QR code + link per kandidat + download ZIP
/admin-event/logout	POST	Logout

3.3 Modul Super Admin

Route	Fungsi
/admin/dashboard	Statistik platform: event, votes, revenue, grafik
/admin/events	CRUD event (nama, org, harga, status, tanggal, icon)
/admin/events/{id}/categories	CRUD kategori dalam event
/admin/events/{id}/teams	CRUD tim peserta dalam event
/admin/accounts	CRUD akun admin event
/admin/transactions	Riwayat semua transaksi + filter
/admin/settings	Pengaturan platform (fee, webhook URL, dll)

4. Rancangan Database (MySQL 8.0)

Semua tabel menggunakan InnoDB engine dengan charset utf8mb4_unicode_ci. Foreign key constraints aktif untuk integritas data.

4.1 Tabel: events

```
CREATE TABLE events (
  id           BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  slug         VARCHAR(120) NOT NULL UNIQUE,
  name         VARCHAR(255) NOT NULL,
  org          VARCHAR(255) NOT NULL,
  description  TEXT NULL,
  icon         VARCHAR(10) DEFAULT '📦',
  price_per_vote INT UNSIGNED NOT NULL DEFAULT 2000,
  min_vote     TINYINT UNSIGNED DEFAULT 1,
  status       ENUM('draft','soon','live','done') DEFAULT 'draft',
  started_at   DATETIME NULL,
  ended_at     DATETIME NULL,
  admin_id     BIGINT UNSIGNED NULL,
  fee_percent  TINYINT UNSIGNED DEFAULT 5,
  created_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,

  INDEX idx_status (status),
  INDEX idx_started (started_at),
  FOREIGN KEY (admin_id) REFERENCES event_admins(id) ON DELETE SET NULL
);
```

4.2 Tabel: categories

```
CREATE TABLE categories (
  id        BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  event_id  BIGINT UNSIGNED NOT NULL,
  name      VARCHAR(100) NOT NULL,
  sort_order TINYINT UNSIGNED DEFAULT 0,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  INDEX idx_event (event_id),
  FOREIGN KEY (event_id) REFERENCES events(id) ON DELETE CASCADE
);
```

4.3 Tabel: teams

```
CREATE TABLE teams (
  id         BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  event_id   BIGINT UNSIGNED NOT NULL,
  category_id BIGINT UNSIGNED NOT NULL,
  number     VARCHAR(5) NOT NULL,         -- '01', '02', dst
  name       VARCHAR(255) NOT NULL,
  location   VARCHAR(255) NULL,           -- kecamatan/kota
  member_count TINYINT UNSIGNED DEFAULT 1,
  avatar_idx TINYINT UNSIGNED DEFAULT 0,  -- 0-9 preset gradient
  vote_count INT UNSIGNED DEFAULT 0,      -- cached counter (PENTING!)
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
  UNIQUE KEY uniq_team_number (event_id, number),
  INDEX idx_event_cat (event_id, category_id),
  INDEX idx_vote_count (event_id, vote_count DESC),
  FOREIGN KEY (event_id) REFERENCES events(id) ON DELETE CASCADE,
  FOREIGN KEY (category_id) REFERENCES categories(id) ON DELETE RESTRICT
);
```

vote_count pada tabel teams adalah denormalized counter untuk performa leaderboard. Selalu update via DB::statement('UPDATE teams SET vote_count = vote_count + ? WHERE id = ?', [$qty, $teamId]) di dalam transaksi.

4.4 Tabel: invoices

```
CREATE TABLE invoices (
  id             BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  xendit_id      VARCHAR(100) NOT NULL UNIQUE,  -- ID dari Xendit
  event_id       BIGINT UNSIGNED NOT NULL,
  team_id        BIGINT UNSIGNED NOT NULL,
  voter_name     VARCHAR(150) NULL,
  vote_qty       TINYINT UNSIGNED NOT NULL DEFAULT 1,
  amount         INT UNSIGNED NOT NULL,          -- dalam Rupiah
  status         ENUM('pending','paid','expired','failed') DEFAULT 'pending',
  paid_at        DATETIME NULL,
  expired_at     DATETIME NULL,
  xendit_payload JSON NULL,                      -- raw webhook payload
  ip_address     VARCHAR(45) NULL,
  created_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,

  INDEX idx_xendit_id (xendit_id),
  INDEX idx_event_status (event_id, status),
  INDEX idx_team_id (team_id),
  INDEX idx_paid_at (paid_at),
  FOREIGN KEY (event_id) REFERENCES events(id),
  FOREIGN KEY (team_id) REFERENCES teams(id)
);
```

Kolom xendit_id adalah kunci idempoten. Gunakan firstOrCreate(['xendit_id' => ...]) untuk mencegah double-count dari webhook duplikat Xendit.

4.5 Tabel: event_admins

```
CREATE TABLE event_admins (
  id         BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name       VARCHAR(150) NOT NULL,
  email      VARCHAR(150) NOT NULL UNIQUE,
  password   VARCHAR(255) NOT NULL,         -- bcrypt
  event_id   BIGINT UNSIGNED NULL,          -- event yang dikelola
  role       ENUM('event_admin','viewer') DEFAULT 'event_admin',
  is_active  BOOLEAN DEFAULT TRUE,
  last_login DATETIME NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,

  INDEX idx_event (event_id),
  FOREIGN KEY (event_id) REFERENCES events(id) ON DELETE SET NULL
);
```

4.6 Tabel: super_admins

```
CREATE TABLE super_admins (
  id         BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name       VARCHAR(150) NOT NULL,
  email      VARCHAR(150) NOT NULL UNIQUE,
  password   VARCHAR(255) NOT NULL,
  is_active  BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

4.7 Tabel: platform_settings

```
CREATE TABLE platform_settings (
  id         BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  key        VARCHAR(100) NOT NULL UNIQUE,
  value      TEXT NULL,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);

-- Seed data default:
INSERT INTO platform_settings (key, value) VALUES
  ('platform_name', 'DigiVote.id'),
  ('fee_percent', '5'),
  ('xendit_webhook_secret', ''),
  ('default_price_per_vote', '2000');
```

Relasi Antar Tabel (ERD Summary)

Tabel	Relasi ke	Tipe	Keterangan
events	event_admins	Belongs To	Satu event dikelola satu admin event
categories	events	Belongs To	Kategori milik satu event
teams	events, categories	Belongs To	Tim masuk satu event & satu kategori
invoices	events, teams	Belongs To	Invoice untuk satu tim di satu event
event_admins	events	Has One	Admin mengelola satu event

5. Optimasi Performa

Semua strategi berikut WAJIB diimplementasi sebelum go-live untuk memastikan 10.000 concurrent user dapat dilayani dengan 200 PHP Worker yang tersedia.

5.1 Konfigurasi .env Wajib

```
# Cache & Session — pakai LSMCD built-in Hostinger
CACHE_DRIVER=memcached
SESSION_DRIVER=memcached
QUEUE_CONNECTION=database

# Memcached (LSMCD default Hostinger)
MEMCACHED_HOST=127.0.0.1
MEMCACHED_PORT=11211
```

5.2 Cache Strategy per Route

Route / Data	Cache Key	TTL	PHP Worker?
Halaman event public	page_{slug}	Auto (LiteSpeed)	✕ 0 — LiteSpeed cache
Data tim per event	teams_{event_id}	3600 detik	☑ Hanya miss pertama
Jumlah vote (leaderboard)	vote_count_{event_id}	30 detik	☑ Hanya miss
Detail event	event_{slug}	600 detik	☑ Hanya miss
Halaman form vote	Tidak di-cache	—	☑ Selalu fresh
Webhook Xendit	Tidak di-cache	—	☑ + Queue dispatch

5.3 Implementasi Cache di Controller

Controller: EventController

```
// Cache data tim — hanya query DB saat cache miss
public function show(string $slug)
{
    $event = Cache::remember('event_'.$slug, 600, fn() =>
        Event::with('categories')->where('slug', $slug)->firstOrFail()
    );

    $teams = Cache::remember('teams_'.$event->id, 3600, fn() =>
        Team::with('category')
            ->where('event_id', $event->id)
            ->orderByDesc('vote_count')
            ->get()
    );

    return view('event.show', compact('event', 'teams'));
}
```

Controller: VoteController

```
// Throttle: max 60 request vote per menit per IP
// (set di routes/web.php: Route::middleware('throttle:60,1'))
```

```
public function store(VoteRequest $request, string $slug)
{
    $event = Cache::remember('event_'.$slug, 600, fn() =>
        Event::where('slug', $slug)->where('status','live')->firstOrFail()
    );

    $team = Team::where('event_id', $event->id)
                ->where('number', $request->team_number)
                ->firstOrFail();

    // Buat invoice Xendit (hit API eksternal)
    $invoice = app(XenditService::class)->createInvoice(
        event: $event,
        team:  $team,
        qty:   $request->vote_qty,
        voter: $request->voter_name
    );

    return response()->json([
        'invoice_url' => $invoice['invoice_url'],
        'xendit_id'   => $invoice['id'],
        'amount'      => $invoice['amount'],
    ]);
}
```

Controller: WebhookController

```
public function xendit(Request $request)
{
    // 1. Validasi signature Xendit
    $secret = config('services.xendit.webhook_secret');
    if ($request->header('x-callback-token') !== $secret) {
        return response()->json(['error' => 'Unauthorized'], 401);
    }

    // 2. Balas Xendit CEPAT — proses di background
    ProcessVotePayment::dispatch($request->all());

    return response()->json(['status' => 'received'], 200);
}
```

Job: ProcessVotePayment

```
class ProcessVotePayment implements ShouldQueue
{
    public function handle(): void
    {
        $data = $this->payload;
        if ($data['status'] !== 'PAID') return;

        // 3. Idempoten — cegah double vote dari webhook duplikat
        $invoice = Invoice::firstOrCreate(
            ['xendit_id' => $data['id']],
            [
                'event_id'   => $this->eventId,
                'team_id'    => $this->teamId,
                'vote_qty'   => $this->qty,
                'amount'     => $data['amount'],
                'status'     => 'paid',
```

```
                'paid_at'    => now(),
                'xendit_payload' => $data,
            ]
        );

        if ($invoice->wasRecentlyCreated) {
            // 4. Update counter di tabel teams (atomic, tidak race condition)
            Team::where('id', $this->teamId)
                ->increment('vote_count', $this->qty);

            // 5. Bust cache vote count agar leaderboard fresh
            Cache::forget('vote_count_'.$this->eventId);
            Cache::forget('teams_'.$this->eventId);
        }
    }
}
```

5.4 Throttle & Rate Limiting

```
// routes/web.php
// Halaman event — bebas (LiteSpeed cache handle)
Route::get('/{slug}', [EventController::class, 'show']);

// Submit vote — throttle ketat
Route::middleware('throttle:60,1')->group(function () {
    Route::post('/vote/{slug}', [VoteController::class, 'store']);
});

// Webhook Xendit — throttle longgar (Xendit server)
Route::middleware('throttle:300,1')->group(function () {
    Route::post('/webhook/xendit', [WebhookController::class, 'xendit']);
});
```

5.5 Optimasi Query Database

Eager Loading — Cegah N+1

```
// ❌ BURUK — N+1 query
$teams = Team::where('event_id', $id)->get();
foreach($teams as $t) echo $t->category->name; // Query per row!

// ✅ BAIK — Eager load
$teams = Team::with('category')->where('event_id', $id)->get();
```

Atomic Increment — Aman untuk Concurrent Update

```
// ✅ Atomic increment (aman saat banyak request bersamaan)
Team::where('id', $teamId)->increment('vote_count', $qty);

// ❌ JANGAN lakukan ini (race condition!)
$team = Team::find($teamId);
$team->vote_count += $qty;
$team->save();
```

Index yang Wajib Ada

Tabel	Kolom Index	Alasan
events	status, started_at	Filter event live/soon di halaman publik
teams	(event_id, vote_count DESC)	ORDER BY untuk leaderboard
teams	(event_id, category_id)	Filter per kategori
invoices	xendit_id	Lookup idempoten di webhook handler
invoices	(event_id, status)	Filter transaksi sukses per event
invoices	paid_at	Filter laporan per tanggal

5.6 Estimasi Kapasitas dengan Strategi Ini

Tipe Request	PHP Worker Dipakai	Keterangan
Buka halaman event (publik)	0	Dilayani LiteSpeed Full Page Cache
Load data tim (cache hit)	0	Served by LSMCD Memcached
Submit form vote (generate invoice)	1 (~300ms)	Hit API Xendit — recycle cepat
Pembayaran QRIS	0	Proses di server Xendit
Webhook konfirmasi bayar	1 (~50ms)	Langsung dispatch Queue, balas cepat
Proses job queue	Background	Tidak memblokir request user
Lihat leaderboard (cache 30 detik)	0 / 1 tiap 30 detik	Cache hit = 0 worker

> *Kesimpulan: Dengan strategi ini, 200 PHP Worker dapat melayani >10.000 concurrent user karena hanya request vote baru yang menyentuh PHP Worker, dan proses tersebut sangat singkat (~300ms).*

6. Integrasi Xendit QRIS

6.1 Flow Pembayaran

Langkah	Pihak	Aksi	Keterangan
1	Frontend	User klik 'Lanjut Bayar'	POST /vote/{slug}
2	VoteController	Panggil XenditService::createInvoice()	Buat invoice di Xendit
3	Xendit	Return invoice_url + QR	Ditampilkan ke user
4	User	Scan QR & bayar di e-wallet	Proses di Xendit
5	Xendit	POST ke /webhook/xendit	Konfirmasi pembayaran
6	WebhookController	Validasi token → dispatch Queue	Balas 200 OK cepat
7	Queue Job	Update vote_count + bust cache	Background process

6.2 XenditService

```
class XenditService
{
    public function createInvoice(Event $event, Team $team, int $qty, ?string
$voter): array
    {
        $amount = $event->price_per_vote * $qty;
        $externalId = 'DV-'.strtoupper($event->slug).'-'.time().'-'.Str::random(6);

        $response = Http::withBasicAuth(config('services.xendit.secret_key'), '')
            ->post('https://api.xendit.co/v2/invoices', [
                'external_id'        => $externalId,
                'amount'             => $amount,
                'description'        => 'Vote '.$qty.'x untuk '.$team->name,
                'invoice_duration'   => 300,   // 5 menit expired
                'payment_methods'    => ['QRIS'],
                'customer'           => ['given_names' => $voter ?? 'Anonymous'],
                'success_redirect_url' => url('/'.$event->slug.'?voted=1'),
                'failure_redirect_url' => url('/'.$event->slug.'?failed=1'),
            ]);

        // Simpan invoice sebagai pending
        Invoice::create([
            'xendit_id' => $response['id'],
            'event_id'  => $event->id,
            'team_id'   => $team->id,
            'vote_qty'  => $qty,
            'amount'    => $amount,
            'status'    => 'pending',
            'expired_at'=> now()->addMinutes(5),
        ]);

        return $response->json();
    }
}
```

6.3 Konfigurasi config/services.php

```
'xendit' => [
    'secret_key'      => env('XENDIT_SECRET_KEY'),
    'webhook_secret'  => env('XENDIT_WEBHOOK_SECRET'),
    'callback_url'    => env('APP_URL').'/webhook/xendit',
],
```

7. Struktur Direktori Laravel

```
app/
├── Http/
│   ├── Controllers/
│   │   ├── EventController.php          // Halaman event publik
│   │   ├── VoteController.php           // Submit vote & generate invoice
│   │   ├── WebhookController.php        // Handler webhook Xendit
│   │   ├── Admin/
│   │   │   ├── DashboardController.php  // Super admin
│   │   │   ├── EventController.php      // CRUD event
│   │   │   └── TransactionController.php
│   │   └── AdminEvent/
│   │       ├── AuthController.php       // Login admin event
│   │       └── DashboardController.php  // Dashboard admin event
│   ├── Middleware/
│   │   ├── SuperAdminAuth.php
│   │   └── EventAdminAuth.php
│   └── Requests/
│       └── VoteRequest.php              // Validasi form vote
├── Jobs/
│   └── ProcessVotePayment.php           // Queue job webhook
├── Models/
│   ├── Event.php
│   ├── Category.php
│   ├── Team.php
│   ├── Invoice.php
│   ├── EventAdmin.php
│   └── SuperAdmin.php
└── Services/
    └── XenditService.php

database/
├── migrations/
│   ├── create_events_table.php
│   ├── create_categories_table.php
│   ├── create_teams_table.php
│   ├── create_invoices_table.php
│   ├── create_event_admins_table.php
│   └── create_super_admins_table.php
└── seeders/
    └── DatabaseSeeder.php

resources/views/
├── event/
│   ├── show.blade.php                   // Halaman publik event
│   └── vote.blade.php                   // Form vote + QRIS
├── admin/                               // Super admin views
└── admin-event/                         // Admin event views
```

8. Prioritas Development & Timeline

Sprint	Durasi	Task	Output
Sprint 1		Setup Laravel, migrasi DB, model & relasi, seeder dummy	Fondasi siap
Sprint 2		Halaman publik event, daftar tim, integrasi LSMCD cache	Voter bisa lihat event
Sprint 3		Form vote, integrasi Xendit, webhook handler, Queue job	Flow voting selesai
Sprint 4		Admin Event: login, dashboard, leaderboard, income, QR download	Admin event siap
Sprint 5		Super Admin: CRUD event/tim/kategori/akun, transaksi	Super admin siap
Sprint 6		Throttle, index DB, load test, optimasi query, LiteSpeed config	Optimasi selesai
Sprint 7		UAT, bug fix, staging deploy ke Hostinger	Go-live ready

8.1 Checklist Sebelum Go-Live

- ☑ CACHE_DRIVER=memcached dan SESSION_DRIVER=memcached aktif di .env production
- ☑ Index database sudah dibuat (terutama invoices.xendit_id UNIQUE)
- ☑ Queue worker aktif dan dimonitor (supervisor atau Hostinger cron)
- ☑ Throttle aktif di route /vote/{slug}
- ☑ Webhook URL sudah didaftarkan di dashboard Xendit
- ☑ Xendit webhook token tersimpan di .env dan divalidasi di WebhookController
- ☑ Invoice::firstOrCreate dengan xendit_id sebagai kunci idempoten
- ☑ LiteSpeed Cache plugin aktif di hPanel Hostinger
- ☑ Load test dengan minimal 500 concurrent request
- ☑ Error logging aktif (Sentry atau Laravel Log Channel)

8.2 Environment Variables Lengkap

```
APP_NAME=DigiVote.id
APP_ENV=production
APP_DEBUG=false
APP_URL=https://digivote.id

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=digivote_db
DB_USERNAME=digivote_user
DB_PASSWORD=your_secure_password

CACHE_DRIVER=memcached
SESSION_DRIVER=memcached
QUEUE_CONNECTION=database
```

```
MEMCACHED_HOST=127.0.0.1
MEMCACHED_PORT=11211

XENDIT_SECRET_KEY=xnd_production_...
XENDIT_WEBHOOK_SECRET=your_webhook_token

MAIL_MAILER=smtp
MAIL_HOST=smtp.hostinger.com
```

Dokumen ini mencakup seluruh fondasi teknis DigiVote.id. Prioritas implementasi adalah: database schema → flow vote+Xendit → caching → admin. Jika ada pertanyaan teknis, diskusikan sebelum mulai sprint terkait.

DigiVote.id · Platform E-Voting Berbayar · DigiTransforma · 2025